

SPA LA SAUVENIERE AP, Belgium

WMO: 064900

Lat: 50.4786N

Lon: 5.9100E

Elev: 482

StdP: 95.67

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-6.4	-14.5	1.1	-4.8	-10.9	1.6	-4.4	11.4	-0.6	10.6	0.7	2.7	150	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	27.8	18.8	25.3	17.7	23.4	16.6	19.6	25.7	18.4	23.8	17.4	22.2	3.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.4	13.2	21.9	16.4	12.4	20.6	15.5	11.7	19.7	57.8	25.9	53.9	24.0	50.8	22.2	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.3	7.5	-10.4	31.9	3.2	1.8	-12.7	33.2	-14.6	34.2	-16.3	35.3	-18.7	36.6		
(5)			-11.1	21.5	3.2	0.8	-13.4	22.1	-15.3	22.6	-17.1	23.0	-19.4	23.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	1.0	1.6	4.6	8.7	12.0	14.8	17.0	16.5	13.1	9.6	5.4	2.5		
	DBStd	6.76	3.78	4.45	4.08	4.28	4.07	3.59	3.73	3.24	3.08	3.82	3.57	3.72		
	HDD10.0	1229	279	237	173	76	26	2	0	0	4	53	144	234		
	HDD18.3	3502	538	470	426	291	201	117	72	76	162	269	389	492		
	CDD10.0	838	0	1	6	36	87	147	217	202	96	42	5	0		
	CDD18.3	69	0	0	0	1	3	12	30	20	4	0	0	0		
	CDH23.3	455	0	0	0	2	16	83	204	133	17	0	0	0		
	CDH26.7	114	0	0	0	0	2	17	59	35	2	0	0	0		
(14) Wind	WSAvg	3.7	4.4	4.2	3.9	3.4	3.2	3.1	3.1	3.3	3.3	3.9	4.3	4.6		
(15) Precipitation	PrecAvg	1042	99	89	77	66	77	82	92	99	80	79	93	111		
	PrecMax	1369	249	228	165	168	182	164	248	230	220	210	161	286		
	PrecMin	772	4	15	10	3	29	25	36	32	9	30	8	18		
	PrecStd	159	51	46	45	36	33	35	43	52	51	37	35	56		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	14.2	17.8	22.7	25.5	29.4	31.7	30.4	26.0	21.4	15.4	11.2		
		MCWB	9.0	6.6	9.8	12.9	16.5	19.6	19.0	19.6	17.7	14.9	12.4	9.4		
	2%	DB	8.8	10.1	15.3	20.2	22.4	25.8	28.0	26.9	22.3	18.6	13.1	9.9		
		MCWB	7.8	5.6	8.7	11.8	14.6	18.2	19.1	18.5	16.0	14.2	10.7	8.5		
	5%	DB	7.2	8.2	12.4	17.8	20.6	23.3	25.4	24.2	20.1	16.4	11.7	8.7		
		MCWB	6.3	6.2	7.7	11.1	14.0	16.7	17.8	17.3	15.3	13.1	10.0	7.6		
	10%	DB	5.9	7.0	10.2	15.4	18.5	21.1	23.2	22.0	17.9	14.7	10.3	7.4		
		MCWB	5.0	5.3	6.7	10.1	13.2	15.7	16.7	16.5	14.2	12.2	9.0	6.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.3	8.5	10.6	13.9	18.0	20.6	20.9	21.0	18.3	16.1	13.3	10.1		
		MCDB	9.9	10.7	15.5	21.1	23.1	27.3	28.4	27.7	24.4	19.5	15.3	10.7		
	2%	WB	7.8	7.5	9.3	12.4	16.5	18.7	19.6	19.2	16.8	14.7	11.2	8.7		
		MCDB	8.5	8.9	13.3	18.4	20.6	24.0	26.1	24.8	20.8	17.5	12.6	9.6		
	5%	WB	6.4	6.5	8.2	11.3	14.7	17.4	18.5	18.0	15.7	13.7	10.2	7.6		
		MCDB	7.2	8.1	11.3	16.6	18.7	22.0	24.2	23.1	19.5	16.0	11.6	8.6		
	10%	WB	5.0	5.5	7.2	10.3	13.5	16.2	17.4	17.0	14.6	12.6	9.0	6.5		
		MCDB	5.8	7.0	9.7	14.9	17.6	20.6	22.3	21.2	17.5	14.5	10.3	7.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.6	4.5	6.1	7.7	7.9	7.9	7.9	7.7	6.6	5.5	4.1	3.8		
		MCDBR	4.6	6.6	9.5	10.3	10.3	11.1	10.7	11.1	9.3	7.5	5.7	4.9		
	5% WB	MCWBR	4.6	4.6	5.2	5.4	5.4	5.8	5.0	5.4	4.9	4.6	4.2	4.1		
		MCWBR	4.6	5.7	7.7	9.4	8.9	9.6	9.7	9.7	8.4	6.8	5.4	4.5		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.322	0.354	0.386	0.394	0.395	0.396	0.389	0.361	0.342	0.317	0.297		
		taud	2.443	2.393	2.333	2.271	2.283	2.295	2.288	2.332	2.395	2.451	2.450	2.421		
	Ebn at Noon		743	813	845	854	862	862	854	842	829	780	711	684		
		Edh at Noon	60	81	102	122	126	125	124	113	96	76	60	53		
(44) All-Sky Solar Radiation	RadAvg		0.77	1.45	2.60	4.12	4.87	5.34	5.07	4.33	3.25	1.91	0.90	0.60		
	RadStd		0.12	0.28	0.38	0.48	0.48	0.50	0.60	0.45	0.37	0.17	0.11	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (77 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page